

The topic is

- A topic is a general subject or theme that can be discussed or written about.
- A topic can be chosen based on the purpose, audience, and context of the communication.
- A topic can be broad or narrow, depending on the scope and depth of the information needed.
- A topic can be expressed as a word, a phrase, a question, or a statement.
- A topic can be derived from various sources, such as personal interests, current events, academic disciplines, or media outlets.
- A topic can be refined or modified by using keywords, subtopics, or aspects to focus on a specific aspect or angle of the topic.
- A topic can be evaluated by considering its relevance, significance, originality, and feasibility for the communication.
- A topic can be developed by researching, brainstorming, outlining, and drafting the content related to the topic.

B.Sc. (Hons.) in Environmental Science

- B.Sc. (Hons.) in Environmental Science is a three-year undergraduate degree course that covers various aspects of the natural and human-made environment, such as ecology, engineering, conservation, biology, and chemistry .
- The course aims to provide students with the knowledge and skills to understand and address the environmental challenges facing humanity, such as climate change, biodiversity loss, pollution, and natural resource management.
- The course consists of 140 credits spread across 26 papers, including eight electives and four ability-enhancement courses. The course also follows the choice based credit system (CBCS) as per the National Education Policy-2020 .
- The course syllabus covers topics such as environmental planning, disaster management, environmental chemistry, environmental biotechnology, environmental law and policy, environmental impact assessment, environmental geology, and environmental economics .
- The course also involves practical sessions, field trips, projects, and internships to provide hands-on experience and exposure to the students .
- The course is accredited by the Institution of Environmental Sciences (IES), which is a professional body that promotes and supports the environmental science profession.
- The course prepares students for various career opportunities in the public and private sectors, such as environmental consultants, environmental engineers, environmental educators, environmental researchers, environ-

mental managers, environmental auditors, and environmental activists .

Unit 1 - Environment: Definition, Types of Environment, Components of environment, Segments of environment, Scope and importance, Need for Public Awareness. Ecosystem: Definition, Types of ecosystem, Structure of ecosystem, Food Chain, Food Web, Ecological pyramid. Balance Ecosystem. Effects of Human Activities such as Food, Shelter, Housing, Agriculture, Industry, Mining, Transportation, Economic and Social security on Environment, Environmental Impact Assessment, Sustainable Development.

- Environment: Definition
 - The environment is the sum of all the physical, chemical and biological factors that surround and influence a living organism.
 - The environment includes both biotic (living) and abiotic (non-living) components.
 - The environment is the area where living organisms live, whereas an ecosystem is a community where the biotic and abiotic elements interact with each other.
- Types of Environment
 - The environment can be classified into two main types: natural and artificial.
 - Natural environment: It refers to the environment that is not modified by human activities, such as forests, oceans, mountains, etc.
 - Artificial environment: It refers to the environment that is created or modified by human activities, such as cities, farms, dams, etc.
- Components of Environment
 - The components of the environment are mainly divided into two categories: biotic and abiotic.
 - Biotic environment: It includes all living organisms such as animals, birds, forests, insects, reptiles and microorganisms like algae, bacteria, fungus, viruses, etc.
 - Abiotic environment: It includes all non-living factors such as air, water, soil, minerals, sunlight, temperature, pressure, etc.
- Segments of Environment
 - The three components of the environment give rise to four important segments of the earth: atmosphere, hydrosphere, lithosphere and biosphere.

- Atmosphere: It is the protective layer of gases that surrounds the earth and sustains life . It absorbs most of the cosmic rays from outer space and regulates the climate.
- Hydrosphere: It is the water component of the earth, which covers about 71% of the earth's surface. It includes oceans, seas, rivers, lakes, glaciers, etc.
- Lithosphere: It is the solid component of the earth, which forms the crust and the upper mantle. It includes rocks, minerals, soil, etc.
- Biosphere: It is the realm of living organisms and their interactions with the atmosphere, hydrosphere and lithosphere. It includes all the ecosystems on the earth.
- Scope and Importance of Environment
 - The scope of environment is very broad, as it covers all the aspects of life on earth.
 - The importance of environment is evident from the fact that it provides the basic necessities for life, such as air, water, food, shelter, etc.
 - The environment also offers various resources for human development, such as energy, materials, medicines, etc.
 - The environment also influences the health, well-being, culture, economy and social security of humans.
 - The environment also plays a vital role in maintaining the ecological balance and biodiversity on the earth.
- Need for Public Awareness
 - The need for public awareness about the environment is crucial, as the human activities have caused various environmental problems, such as pollution, deforestation, climate change, loss of biodiversity, etc.
 - The public awareness can help in creating a sense of responsibility and respect for the environment among the people.
 - The public awareness can also help in promoting the conservation and sustainable use of the environmental resources.
 - The public awareness can also help in influencing the policies and actions of the government and other stakeholders for the protection and improvement of the environment.
- Ecosystem: Definition
 - An ecosystem is a functional unit of nature, where living organisms interact with each other and with the abiotic environment.
 - An ecosystem is a dynamic system, where energy and matter flow and cycle among the biotic and abiotic components.
 - An ecosystem can be of any size, from a drop of water to the whole biosphere.

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Types of Ecosystem

Unit 2 - Natural Resources: Introduction, Classification. Water Resources; Availability, sources and Quality Aspects, Water Borne and Water Induced Diseases, Fluoride and Arsenic Problems in Drinking Water. Mineral Resources; Material Cycles; Carbon, Nitrogen and Sulfur cycles. Energy Resources; Conventional and Non conventional Sources of Energy. Forest Resources; Availability, Depletion of Forests, Environment impact of forest depletion on society.

- **Introduction:** Natural resources are substances or materials from nature that humans use to sustain life. These resources are naturally available on Earth without any human action. Natural resources have economic, ecological, scientific, aesthetic, and cultural values in human life. Some examples of natural resources are oil, coal, natural gas, metals, stone, sand, air, sunlight, soil, and water.
- **Classification:** Natural resources can be classified into different categories based on their origin, level of development and uses, stock or deposits, and distribution. On the basis of their origin, natural resources can be classified into living or biotic and non-living or abiotic resources. Living or biotic resources are derived from living things or organic materials, such as plants, animals, and microorganisms. Non-living or abiotic resources are derived from nonliving or inorganic materials, such as minerals, metals, and fossil fuels. On the basis of their level of development and uses, natural resources can be classified into renewable and non-renewable resources. Renewable resources are the resources that can be regenerated or replenished by natural processes within a reasonable time, such as sunlight, water, and wood. Non-renewable resources are the resources that cannot be regenerated or replenished by natural processes within a reasonable time, such as coal, oil, and natural gas. On the basis of their stock or deposits, natural resources can be classified into actual and potential resources. Actual resources are the resources that have been surveyed, quantified, and are being used, such as coal mines, oil wells, and hydroelectric dams. Potential resources are the resources that have not been surveyed, quantified, or used, but are expected to exist and have some utility in the future, such as uranium deposits, geothermal energy, and tidal energy. On the basis of their distribution, natural resources can be classified into ubiquitous and localized resources. Ubiquitous resources are the resources that are found everywhere or in most places on Earth, such as air, sunlight, and soil. Localized resources are the resources that are found only in certain regions or places on Earth, such as coal, oil, and

minerals.

- **Water Resources:** Water is one of the most essential and abundant natural resources on Earth. It covers about 71% of the Earth's surface and is vital for all forms of life. Water resources are the sources of water that are useful or potentially useful for human use. Water resources can be classified into surface water and groundwater. Surface water is the water that is found on the Earth's surface, such as rivers, lakes, ponds, streams, and oceans. Groundwater is the water that is stored underground in aquifers, which are porous rocks or sediments that can hold and transmit water. Water resources can be used for various purposes, such as drinking, irrigation, sanitation, hygiene, industry, recreation, and power generation. However, water resources are also facing many challenges, such as availability, quality, and pollution.
 - **Availability:** The availability of water resources depends on the quantity and distribution of water on Earth. Although water is abundant on Earth, only about 2.5% of it is fresh water, and only about 0.3% of it is accessible for human use. The rest of the water is either saline, frozen, or deep underground. Moreover, the distribution of water resources is uneven across the world, with some regions having more water than others. For example, Asia has the largest amount of freshwater resources, but also the largest population and water demand. Africa has the lowest amount of freshwater resources, but also the highest rate of water scarcity and stress. Therefore, the availability of water resources is a major concern for many countries and regions, especially in the face of climate change, population growth, and urbanization[[^]

Unit 3 - Pollution and their Effects; Public Health Aspects of Environmental; Water Pollution, Air Pollution, Soil Pollution, Noise Pollution, Solid Waste Management

- Pollution is the introduction of harmful substances or energy into the environment that cause adverse effects on living organisms and natural resources.
- Pollution can be classified into different types based on the source, the medium, and the effect of the pollutants. Some of the common types of pollution are water pollution, air pollution, soil pollution, noise pollution, and solid waste management.
- Water pollution is the contamination of water bodies such as rivers, lakes, oceans, and groundwater by human activities or natural processes. Water pollution can affect the quality and availability of water for drinking, irrigation, recreation, and ecosystem services. Some of the causes of water pollution are industrial effluents, agricultural runoff, sewage disposal, oil spills, mining, and thermal pollution.

- Air pollution is the presence of harmful substances or particles in the atmosphere that can affect the health and well-being of humans and other living organisms. Air pollution can also cause climate change, acid rain, smog, and ozone depletion. Some of the sources of air pollution are fossil fuel combustion, vehicle emissions, biomass burning, volcanic eruptions, and dust storms.
- Soil pollution is the degradation of soil quality and fertility due to the accumulation of harmful substances or changes in the natural properties of the soil. Soil pollution can affect the productivity and diversity of crops and plants, and also contaminate the food chain and water resources. Some of the factors that contribute to soil pollution are over-irrigation, overuse of pesticides and fertilizers, dumping of solid waste, deforestation, and mining .
- Noise pollution is the excessive or unwanted sound that can cause annoyance, stress, hearing loss, and other health problems for humans and animals. Noise pollution can also interfere with communication, learning, and work performance. Some of the sources of noise pollution are traffic, construction, industrial machinery, aircraft, fireworks, and loud music.
- Solid waste management is the collection, transportation, treatment, and disposal of solid waste generated by human activities or natural processes. Solid waste can be classified into different types based on the origin, composition, and biodegradability of the waste. Some of the types of solid waste are municipal solid waste, industrial waste, hazardous waste, biomedical waste, and electronic waste. Solid waste management aims to reduce the environmental and health impacts of solid waste by applying the principles of reduce, reuse, recycle, and recover.
- Pollution and its effects have significant implications for public health and environmental sustainability. Pollution can cause various diseases and disorders, such as respiratory infections, asthma, cancer, cardiovascular diseases, diabetes, and premature death . Pollution can also damage the natural resources and ecosystems that support life on Earth, such as water, soil, air, biodiversity, and climate. Pollution can also affect the social and economic aspects of human development, such as poverty, inequality, education, and human rights.
- To prevent and control pollution, various measures and strategies can be adopted at different levels, such as individual, community, national, and international. Some of the measures and strategies are:
 - Adopting cleaner and renewable sources of energy, such as solar, wind, and hydro power.
 - Implementing stricter regulations and standards for emission and discharge of pollutants, and enforcing compliance and accountability.
 - Promoting public awareness and education on the causes and effects of pollution, and encouraging behavioral change and civic responsi-

bility.

- Developing and applying innovative technologies and solutions for pollution prevention, monitoring, and remediation.
- Enhancing cooperation and coordination among different stakeholders, such as governments, industries, civil society, and international organizations, to address the global and transboundary challenges of pollution.

Unit 4 - Current Environmental Issues of Importance

- **Global warming** is the rise in the average temperature of the Earth's climate system due to the increased concentration of greenhouse gases in the atmosphere, mainly from human activities such as burning fossil fuels, deforestation, and agriculture. Global warming causes many negative impacts on the environment, such as melting ice caps, sea level rise, extreme weather events, droughts, floods, wildfires, biodiversity loss, and spread of diseases .
- **Greenhouse effect** is the natural process by which the Earth's atmosphere traps some of the outgoing heat from the sun and keeps the planet warm enough to sustain life. Without the greenhouse effect, the Earth would be too cold for most living things. However, human activities have enhanced the greenhouse effect by adding more greenhouse gases, such as carbon dioxide, methane, nitrous oxide, and chlorofluorocarbons, to the atmosphere, which trap more heat and cause global warming .
- **Climate change** is the long-term alteration of the Earth's climate system, including changes in temperature, precipitation, wind patterns, ocean currents, and other aspects of the weather. Climate change can be natural or human-induced, but the current climate change is largely driven by human activities that increase greenhouse gas emissions. Climate change affects the environment, society, and economy in various ways, such as altering ecosystems, threatening food security, increasing health risks, displacing populations, and disrupting infrastructure .
- **Acid rain** is the precipitation that contains high levels of acidic pollutants, such as sulfur dioxide and nitrogen oxides, which are emitted from coal-fired power plants, factories, and vehicles. Acid rain can damage buildings, monuments, statues, and other structures made of stone, metal, or concrete. It can also harm forests, crops, soils, lakes, rivers, and aquatic life by lowering the pH and leaching nutrients and metals .
- **Ozone layer** is the layer of the Earth's stratosphere that contains a high concentration of ozone, a gas that absorbs most of the harmful ultraviolet radiation from the sun and protects life on Earth from skin cancer, eye damage, and immune system disorders. However, human activities have caused the depletion of the ozone layer by releasing ozone-depleting substances, such as chlorofluorocarbons, halons, and methyl bromide, which break down ozone molecules in a chemical reaction. The ozone layer depletion has resulted in the formation of the ozone hole, a region of low

ozone concentration over the Antarctic and Arctic regions, which allows more ultraviolet radiation to reach the Earth's surface .

- **Population growth** is the increase in the number of people living on the planet, which has reached about 7.9 billion as of 2023. Population growth puts pressure on the environment, as more people consume more resources, produce more waste, and occupy more land. Population growth also contributes to other environmental problems, such as global warming, greenhouse effect, climate change, acid rain, ozone layer depletion, deforestation, biodiversity loss, water scarcity, soil erosion, and pollution .
- **Automobile pollution** is the emission of harmful substances, such as carbon monoxide, nitrogen oxides, sulfur dioxide, particulate matter, and volatile organic compounds, from the exhaust of vehicles that use gasoline, diesel, or other fossil fuels. Automobile pollution can affect the air quality, human health, and the environment. Automobile pollution can cause respiratory problems, cardiovascular diseases, cancer, and premature death in humans. It can also contribute to global warming, greenhouse effect, climate change, acid rain, ozone layer depletion, smog, and visibility reduction .
- **Burning of paddy straw** is the practice of setting fire to the crop residue left after harvesting rice, wheat, or other grains. Burning of paddy straw is mainly done by farmers to clear the field quickly and cheaply for the next crop, or to get rid of pests and diseases. However, burning of paddy straw causes air pollution, greenhouse gas emissions, soil degradation, and health hazards. Burning of paddy straw releases large amounts of smoke, ash, and carbon dioxide, which can worsen the air quality, aggravate respiratory and cardiovascular diseases, and contribute to global warming, greenhouse effect, climate change, and acid rain^[1]

Unit 5 - Environmental Protection

- Environmental Protection Act 1986
 - An Act of the Parliament of India that came into force on 19 November 1986 .
 - It has 26 sections and 4 chapters .
 - It is widely considered to have been a response to the Bhopal gas leak .
 - It empowers the Central Government to establish authorities, issue rules and directions, and plan and execute a nation-wide programme for the prevention, control and abatement of environmental pollution .
 - It also provides for penalties and procedures for handling violations of the Act or the rules made thereunder.
- Initiatives by Non Governmental Organizations (NGO's)
 - NGO's are voluntary organizations that work for various social, environmental, and developmental causes.

- Some examples of NGO's working for environmental protection in India are:
 - * Centre for Science and Environment (CSE): It is a public interest research and advocacy organization that promotes environmentally sound and equitable development policies.
 - * World Wide Fund for Nature (WWF): It is a global conservation organization that works to protect biodiversity, reduce ecological footprint, and promote sustainable development.
 - * Greenpeace India: It is a branch of the international environmental organization that campaigns for clean energy, forest protection, sustainable agriculture, and climate justice.
 - * Wildlife Trust of India (WTI): It is a conservation organization that works to protect wildlife and habitats, mitigate human-wildlife conflicts, and support community-based conservation initiatives.
 - * Bombay Natural History Society (BNHS): It is one of the oldest scientific organizations in India that conducts research, education, and advocacy on natural history, biodiversity, and conservation.
- Human Population and the Environment
 - Population growth
 - * Population growth is the increase in the number of people in a given area or region over time.
 - * Population growth can have positive and negative impacts on the environment.
 - * Positive impacts include increased human capital, innovation, and economic development.
 - * Negative impacts include increased consumption of natural resources, waste generation, pollution, deforestation, habitat loss, and climate change.
 - Environmental Education
 - * Environmental education is the process of learning about the natural and human-made environment and developing skills, attitudes, and values to protect and improve it.
 - * Environmental education can help to raise awareness, foster positive attitudes, and promote responsible actions towards the environment.
 - * Environmental education can be formal, non-formal, or informal.
 - * Formal environmental education is integrated into the curriculum and taught in schools and colleges.
 - * Non-formal environmental education is delivered through various programs and activities by NGOs, media, government agencies, and other institutions.
 - * Informal environmental education is acquired through everyday experiences, interactions, and observations of the environment.
 - Women Education

- * Women education is the process of providing equal access and opportunity to education for women and girls.
- * Women education can have positive impacts on the environment.
- * Women education can help to reduce population growth, improve maternal and child health, enhance women's empowerment and participation, and foster environmental awareness and stewardship.